



On Quantified Observability Analysis in Multi-Agent Systems

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Outline



- 1 Motivations and overview
- 2 Modelling POMASs
- 3 Observability Specification: the logic oPATL
- 4 Verification of Observability Properties
- 5 Conclusions

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Motivations

- In MASs, agents' observability on system behaviours may improve the overall team performance, but
 - ▶ may also release sensitive information to malicious (inside/outside) agents and cause information leakage.
- Information exposure issue should also play a role in making decisions for agents.
- A **quantified observability** analyses can thus be useful in MASs to assist decision-making in MASs design to address such concerns.



What do we do?

- A framework for specification and verification of quantified observability properties in MASs:
 - ▶ The model: Partially Observable multi-player Markov chain game
 - ▶ Specification: the logic of oPATL, to express (probabilistic) observability properties
 - ▶ Verification: probabilistic model checking, to allow automatically verify quantified quantified observability properties in MASs



What do we do?

- A framework for specification and verification of quantified observability properties in MASs:
 - ▶ The model: Partially Observable multi-player Markov chain game
 - ▶ Specification: the logic of oPATL, to express (probabilistic) observability properties
 - ▶ Verification: probabilistic model checking, to allow automatically verify quantified quantified observability properties in MASs
- Implementation and case studies:
 - ▶ prototype: built upon the PRISM model checker
 - ▶ examples: voting, supply-chain, p2p-network

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Definition: probabilistic game structure

A PGS is a tuple $\mathcal{G} = (S, \text{Act}, \delta)$, where:

- S is a finite set of *states*;
- $\text{Act} = \prod_{j \in \text{Ag}} \text{Act}_j$ is a finite set of joint actions (decisions) of the agents in Ag , $\text{Act}_j \subseteq \Sigma$ is the set of actions that $j \in \text{Ag}$ can perform;
- $\delta : S \rightarrow 2^{\text{Dist}(\text{Act} \times S)}$ is the *probabilistic transition relation*; for state $s \in S$, $\delta(s)$ is the distribution for next state; s is a *terminal state* if $\delta(s) = \emptyset$.



Definition: Observations

- Let Θ be a finite alphabet for observables, and $\Theta^\epsilon = obs \cup \{\epsilon\}$ where ϵ denotes the invisible/hidden action.
- An **observation function** on paths is a labelled-based function

$$obs : \text{Paths}_{\mathcal{G}}(s) \rightarrow (\Theta_1 \times \Theta_2 \times \cdots \times \Theta_n)^\infty$$

where $\Theta_j \subseteq \Theta$ denotes a finite set of observables for $j \in \text{Ag}$.



Definition: POMAS

A POMAS is a tuple $\mathcal{M} = (\text{Ag}, \mathcal{G}, s_0, \text{Ap}, L, \{\text{obs}_i\}_{i \in \text{Ag}})$, where:

- $\text{Ag} = \{1, \dots, n\}$ is a finite set of intelligent *agents*;
- $\mathcal{G} = (S, \text{Act}, \delta)$ is a *fully PGS*;
- $s_0 \in S$ is the *initial state*;
- Ap is a finite set of *atomic propositions*;
- $L : S \rightarrow 2^{\text{Ap}}$ is the state labelling function mapping each state to a set of atomic state proposition taken from set Ap ;
- $\text{obs}_i : \text{Paths}_{\mathcal{G}}(s_0) \rightarrow (\Theta_1 \times \dots \times \Theta_n)^\infty$ is an observation function for agent $i \in \text{Ag}$.



Example: Voting

1. The chair initiates the voting procedure;
2. Voters simultaneously propose their votes;
3. Each voter commits his vote once he makes his decision;
4. If a voter is waiting during this process, he can partially observe the behaviour of other voters and gather indications, such as identifying the dominant candidate;
5. Based on their observations, the voter can make their own voting decision;
6. Once all voters have committed their votes, the chair closes the voting session.



Example: Voting

Actions	$\zeta_1(\text{Action})$	$\zeta_2(\text{Action})$	$\zeta_3(\text{Action})$	Descriptions
Op_0	Op_0	Op_0	Op_0	the chair opens a voting session
Cl_0	Cl_0	Cl_0	Cl_0	the chair closes a voting session
W_i	W	W	W	agent i is waiting, $i \in \{0, 1, 2, 3\}$
V_1^X	X_1	ϵ	X	voter 1 votes candidate X
V_1^Y	Y_1	ϵ	Y	voter 1 votes candidate Y
V_2^X	ϵ	X_2	X	voter 2 votes candidate X
V_2^Y	ϵ	Y_2	Y	voter 2 votes candidate Y
V_3^X	ϵ	ϵ	X_3	voter 3 votes candidate X
V_3^Y	ϵ	ϵ	Y_3	voter 3 votes candidate Y



Example: Voting

- Assume voter 2 is the observer, the actions of voting X by 1 and 3 are not visible to voter 2.
- Consider a path:

$$\rho = s_0 \xrightarrow{Op_0 W_1 W_2 W_3} s_1 \xrightarrow{\frac{1}{2} \cdot W_0 V_1^X W_2 W_3} s_2 \xrightarrow{\frac{1}{2} \cdot W_0 W_1 V_2^Y W_3} s_3 \xrightarrow{\frac{2}{3} \cdot W_0 W_1 W_2 V_3^Y} s_4 \xrightarrow{Cl_0 W_1 W_2 W_3} s_5$$

- The observation from 2's view:

$$obs_2(\rho) = Op_0 WWW WWY_2W WWW Cl_0WWW \text{ w.p. } \frac{1}{6}.$$



Definition: Mixed Strategy

A *mixed strategy* for agent i is a function π_i :

$$\pi_i \triangleq \text{obs}_i(\text{Paths}_{\mathcal{G}}(s_0)) \rightarrow \text{Dist}(\text{Act}_i)$$

such that, if $\pi_i(\rho)(a_i) > 0$ then $a_i \in \text{Act}_i(\text{last}(\rho))$. The set of all strategies of agent i is denoted Π_i .

Definition: Strategy profile

A *strategy profile* for POMAS $\mathcal{M}$ is a tuple

$$\pi = (\pi_1, \dots, \pi_n) \in \Pi_1 \times \dots \times \Pi_n$$

producing a strategy for each agent of the system.



Example: strategy in voting

- Assume voter 3 is the observer, and consider the path:

$$\rho = s_0 \xrightarrow{Op_0 W_1 W_2 W_3} s_1 \xrightarrow{\frac{1}{4} \cdot W_0 V_1^X V_2^Y W_3} s_2 \xrightarrow{\frac{1}{3} \cdot W_0 W_1 W_2 V_3^X} s_3 \xrightarrow{Cl_0 W_1 W_2 W_3} s_4$$



Example: strategy in voting

- Assume voter 3 is the observer, and consider the path:

$$\rho = s_0 \xrightarrow{0p_0w_1w_2w_3} s_1 \xrightarrow{\frac{1}{4} \cdot w_0 V_1^X V_2^Y w_3} s_2 \xrightarrow{\frac{1}{3} \cdot w_0 w_1 w_2 V_3^X} s_3 \xrightarrow{Cl_0 w_1 w_2 w_3} s_4$$

- The observation on the above path from voter 3's view would then be:
 $obs_3(\rho) = 0p_0WWW WXYW WWWX_3 Cl_0WWW \text{ w.p. } \frac{1}{12}.$



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- The observation on the above path from voter 3's view would then be:
 $obs_3(\rho) = 0p_0WWW WXYW WWWX_3 Cl_0WWW$ w.p. $\frac{1}{12}$.
- The observer's information of knowledge obtained from her observation might influence her decision on voting.
- Assume the observer (voter 3) is not able to see whom other voters have voted, but she can see how many ballots each candidate has received
- A basic strategy to reflect such an influence is that:
 - she will vote the dominant candidate if there is one, otherwise
 - she will vote the candidates under her preferred distribution.

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Observability and Opacity

Let $\mathcal{M} = (\text{Ag}, \mathcal{G}, s_0, \text{Ap}, L, \{\text{obs}_i\}_{i \in \text{Ag}})$. Given a predicate φ over $\text{Paths}_{\mathcal{G}}(s_0)$, we say φ is **opaque** w.r.t. obs_i if:

- for every path $\rho \in \llbracket \varphi \rrbracket$, there is a path $\rho' \in \llbracket \bar{\varphi} \rrbracket$ s.t.

$$\text{obs}(\rho) = \text{obs}(\rho')$$

- i.e., under obs_i , all paths satisfying φ are covered by paths in $\llbracket \bar{\varphi} \rrbracket$:

$$\text{obs}_i(\llbracket \varphi \rrbracket) \subseteq \text{obs}_i(\llbracket \bar{\varphi} \rrbracket)$$

where $\llbracket \bar{\varphi} \rrbracket \triangleq \text{Paths}_{\mathcal{G}}(s_0) \setminus \llbracket \varphi \rrbracket$.



The logic oPATL: syntax

Three classes of formulae: **state formulae** ϕ , **path formulae** ψ , and **observability formulae** Φ :

$$\begin{aligned}\phi &::= a \mid \neg\phi \mid \phi \wedge \phi \mid \mathbf{P}_{\bowtie p} \langle A \rangle [\psi] \mid \mathcal{D}_{\bowtie p} \langle A \rangle [\Phi] \\ \psi &::= \mathbf{X}\phi \mid \phi \mathbf{U}\phi \mid \phi \mathbf{R}\phi \mid \neg\psi \mid \psi \wedge \psi \\ \Phi &::= \mathbf{O}_i [\psi] \mid \neg\Phi \mid \Phi \wedge \Phi\end{aligned}$$



Voting Example: observability

- Consider the property ψ : “eventually candidate X wins”
- The probabilistic observability property of voter 2:

$$\mathcal{D}_{\leq \frac{1}{3}} \langle 1, 2, 3 \rangle [\mathbf{O}_2[\psi]] = \text{true}$$

since:

$$\begin{aligned} & \text{Prob}(\llbracket \langle 1, 2, 3 \rangle \mathbf{O}_2[\psi] \rrbracket) \\ &= \text{Prob}(\{ \frac{1}{12} \cdot \text{op}_0 V_1^X V_2^X V_3^X \text{cl}_0, \frac{1}{6} \cdot \text{op}_0 V_1^X V_2^X V_3^Y \text{cl}_0 \}) = \frac{1}{4} < \frac{1}{3} \end{aligned}$$

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Verification of $\mathcal{D}_{\bowtie p}[\mathbf{O}_i \langle A \rangle \psi]$

Given a POMAS $\mathcal{M} = (\mathcal{G}, s_0, \text{Ag}, \text{Ap}, \{\text{obs}_i\}_{i \in \text{Ag}})$, a task in property ψ required to be completed under a strategy π_A of a coalition $A \in \text{Ag}$, the probabilistic verification problem of observability property is to decide whether $s_0 \models_{\mathcal{M}} \mathcal{D}_{\bowtie p}(\mathbf{O}_i \langle A \rangle [\psi])$, i.e.,

$$\mathbb{P}_{s_0}(\llbracket \langle A \rangle [\psi] \rrbracket \setminus \text{obs}_i^{-1}(\text{obs}_i(\llbracket \langle A \rangle [\neg \psi] \rrbracket))) \bowtie p.$$



Checking $s \models_{\mathcal{M}} \mathcal{D}_{\bowtie p}(\mathbf{O}_i \langle \mathbf{A} \rangle [\psi])$

- Find all consistent paths Π and the corresponding traces Λ , satisfying ψ under mixed strategy π_A of coalition A:

$$\Pi = \{\text{Reg}(\rho_{\pi_A}) \mid \rho_{\pi_A} \models_{\mathcal{M}} \psi\} \quad \Lambda = \{\text{erase}(\rho) \mid \rho \in \Pi\}.$$



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- Find all consistent paths Π' and the corresponding traces Λ , violating ψ under mixed strategy π_A of coalition A:
 $\Pi' = \{\text{Reg}(\rho'_{\pi_A}) \mid \rho_{\pi_A} \not\models_{\mathcal{M}} \psi\} \quad \Lambda' = \{\text{erase}(\rho') \mid \rho' \in \Pi'\}.$



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- Find all ψ -opaque traces:
 $\Lambda'' = \{\lambda'' \mid \lambda'' \in \Lambda \wedge \exists \lambda' \in \Lambda'. (\text{obs}_i(\lambda') = \text{obs}_i(\lambda''))\}.$



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- Compute the probability of ψ -observable traces:
 $d = \mathbb{P}_{s_0}(\llbracket \mathbf{O}_i \langle A \rangle [\psi] \rrbracket) = \sum_{\xi \in (\Lambda \setminus \Lambda'')} \text{Prob}(\xi).$



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- Return true if $d \bowtie p$, return false otherwise.



Voting Example: observability

Probabilistic verification of $\mathcal{D}_{=?}\langle A \rangle [\mathbf{O}_i[\psi]]$

Result: 0.25.{

0.083333333333333333:vX1vX2vX3c10→X2c10,

0.166666666666666666:vX1vX2vY3c10→X2c10

} (value in the initial state)

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Conclusions

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- Applications: information security analysis in MASs



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Future work

- Integration of the observability operator into **Strategic Logic** and evaluation its suitability for various scenarios
- Trade-off: performance vs. information leakage/observability
 - ▶ e.g., game-theoretic methods

Thank You! ☺